



Fig. 3: Crash tests (interferences)

of a shaft and the length of the shaft is changed, the groove remains in constant distance from the end edge. Constraints can be defined within a 2D profile creating a solid (for instance, by extrusion), among elements of a solid or among entire solids.

Whenever the user enters a dimension in 2D profile used for solid creation, a distance in constraint or a dimension of solid, it is possible to optionally use parameter or even a mathematical expression containing parameters. One can change shapes or locations of solids by changing parameter values.

3D assemblies and groups. VariCAD also provides assembly support tools. If the link between a part and assembly is defined, any changes to be made to the part file are reflected in the assembly file and vice-versa. Linked copies of solids (also known as identical solids) can also be defined.

When creating assemblies, it is up to the user whether he wants to use links between the assembly and parts or have all-in-one document. If you decide to use these links, all the changes in assemblies or parts are automatically made also in all the drawings you need.

Crash tests (interferences). Component interference checking is one of the excellent features of 3D modeling. VariCAD can check 3D assemblies for possible collisions (overlapping vol-

ume) between components.

Calculations. Moment of inertia, centre of gravity, mass, volume, 2D section area and surface area can be calculated with the help of VariCAD. Mechanical parts calculations are also included for standard parts used everyday by mechanical designers. There are calculations of belt drives, pre-stressed bolted connections, spur, grooved shafts, pins and parallel keys, bearings, tension and compression springs, beams under combined stress (bending and torsion) and bevel gearing geometry.

VariCAD contains many tools for checking the created models. It detects interferences between solids, and checks attributes of the solids so that these are correctly inserted into BOM or your corporate information system.

Surface development (sheet metal unbending). VariCAD can also create developed (unbent) surfaces of sheet metal parts or 3D solids. For further processing, the XY coordinates of developed surfaces can be saved into a text file. The user can enter bending coefficients in order to customise his calculations to reflect material and technology.

Mechanical part and symbol libraries. VariCAD contains both 3D and 2D libraries of mechanical parts (ANSI, DIN) such as bearings, nuts, cotter pins to name a few, hydraulic and

Tools within VariCAD

- 3D modeling
- 2D drawing and editing
- Optional support of parameters
- Optional support of geometrical constraints
- Shells modeling, pipelines, wires
- Crash tests (interferences)
- 3D assemblies and groups
- Surface development (sheet metal unbending)
- Mechanical part libraries and symbol libraries
- Calculations of 3D objects or 2D sections
- BOM and title blocks

Minimum system requirements for VariCAD

Windows version

- At least 2GHz processor
- 2GB RAM
- 250MB HDD
- Graphic card with hardware support for OpenGL graphics
- Three-button mouse
- Microsoft Windows XP (SP2), Windows Vista, Windows 7 or Windows 8

Linux version

- OpenGL libraries
- XServer supporting OpenGL (extension GLX)
- At least 2 GHz
- 1GB RAM, 250MB HDD
- Graphic card with hardware support for OpenGL graphics
- For VariCAD 64-bit, more memory recommended than for a 32-bit version

Note. True Colour mode and hardware support of OpenGL are always required.

What's New in VariCAD 1.03 (as on January 30, 2013)

- Changes in 3D kernel, allowing to read multiple shape representations from STEP
- Support of AutoCAD 2013's DWG/DXF format
- Tools for working with open solids (mostly loaded from STEP)
- New repair tools that are accessible from pull-down menu or from a new toolbar
- Completely rebuilt working with construction lines in 2D
- Minor improvements in 3D kernel and few changes in the user interface